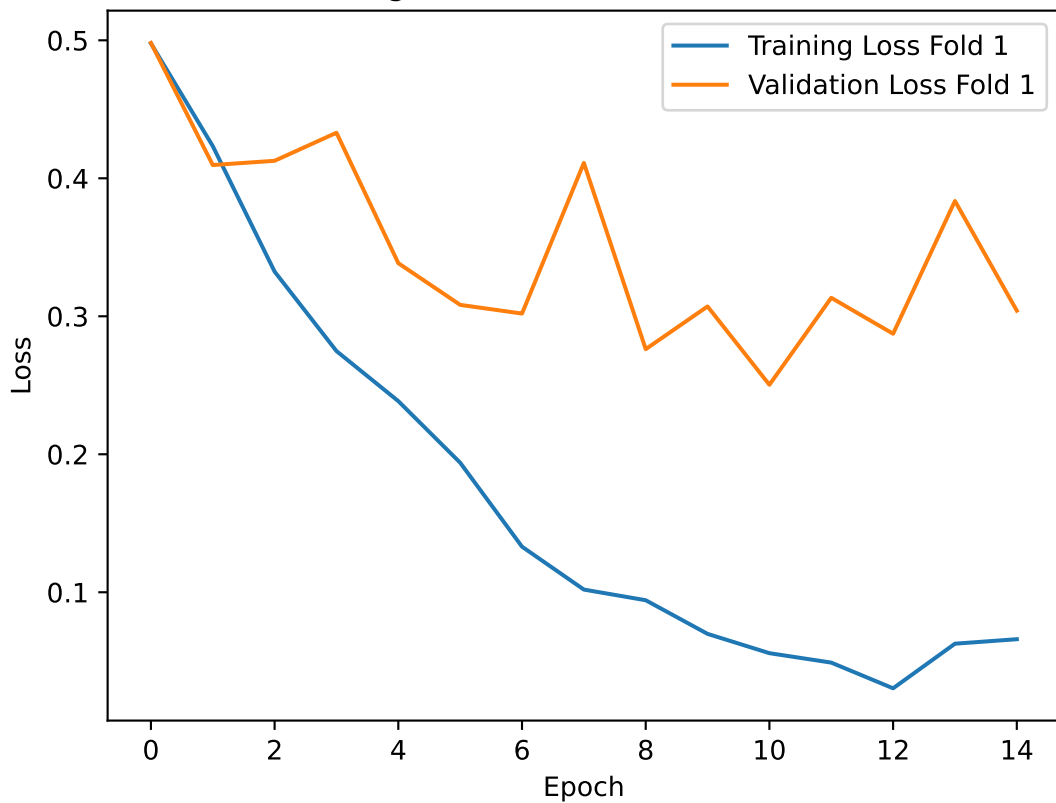


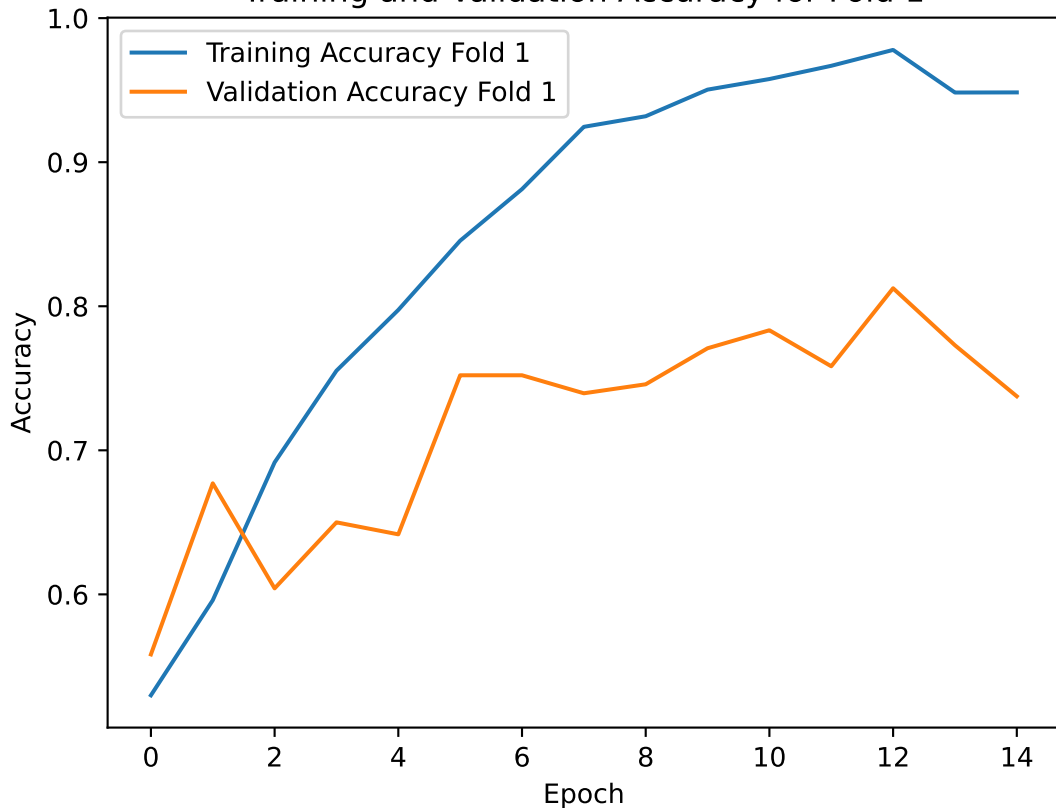
Validation F1-Score for fold 1: 0.1631205673758865

0.1631205673758865

Training and Validation Loss for Fold 1



Training and Validation Accuracy for Fold 1

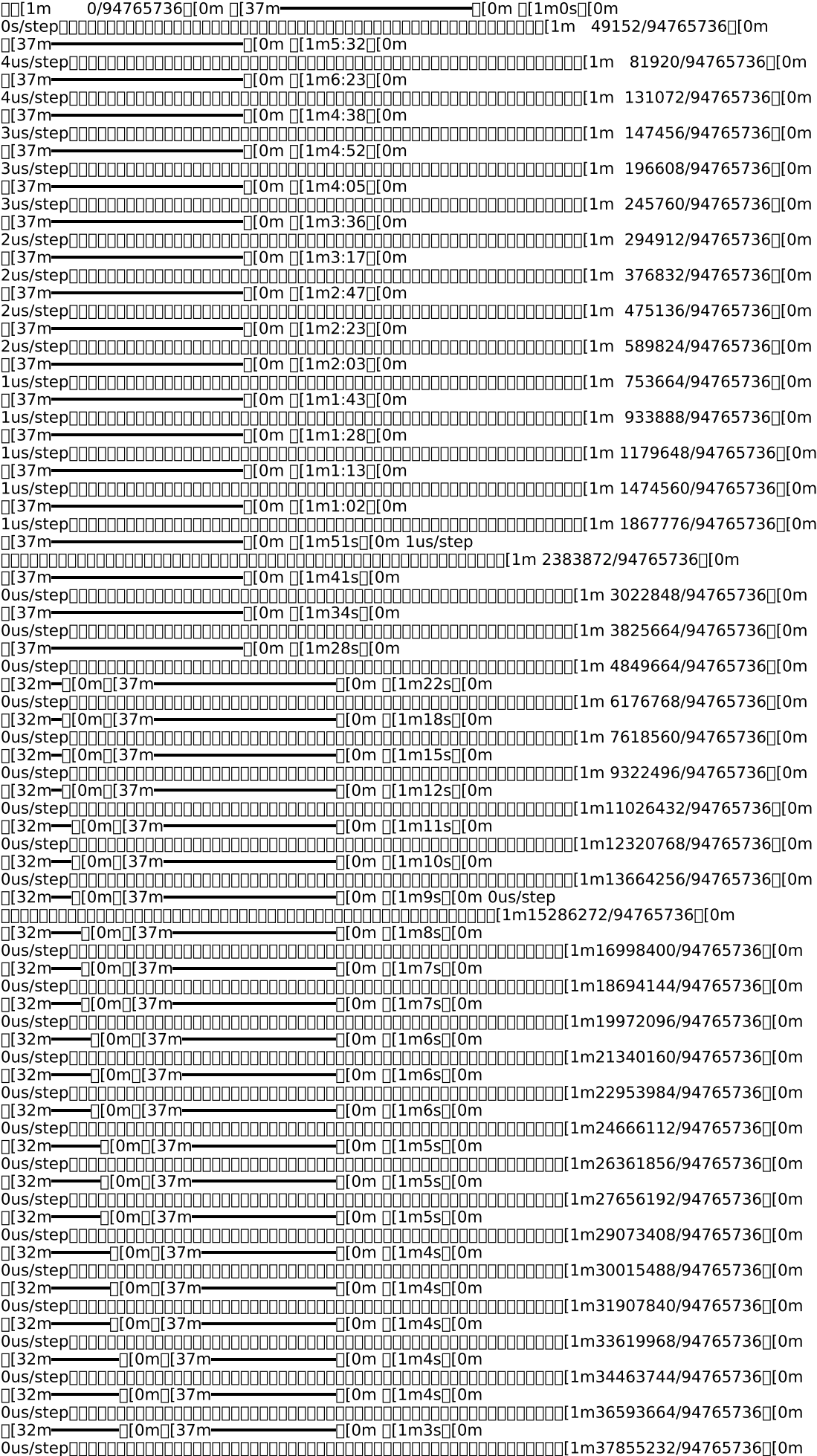


Best Hyperparameters for Fold 1:

`{'conv_filters': 64, 'conv_activation': 'relu', 'unitsD1': 288, 'unitsD2': 160}`

Best Trial Score for Fold 1: 0.8166667222976685

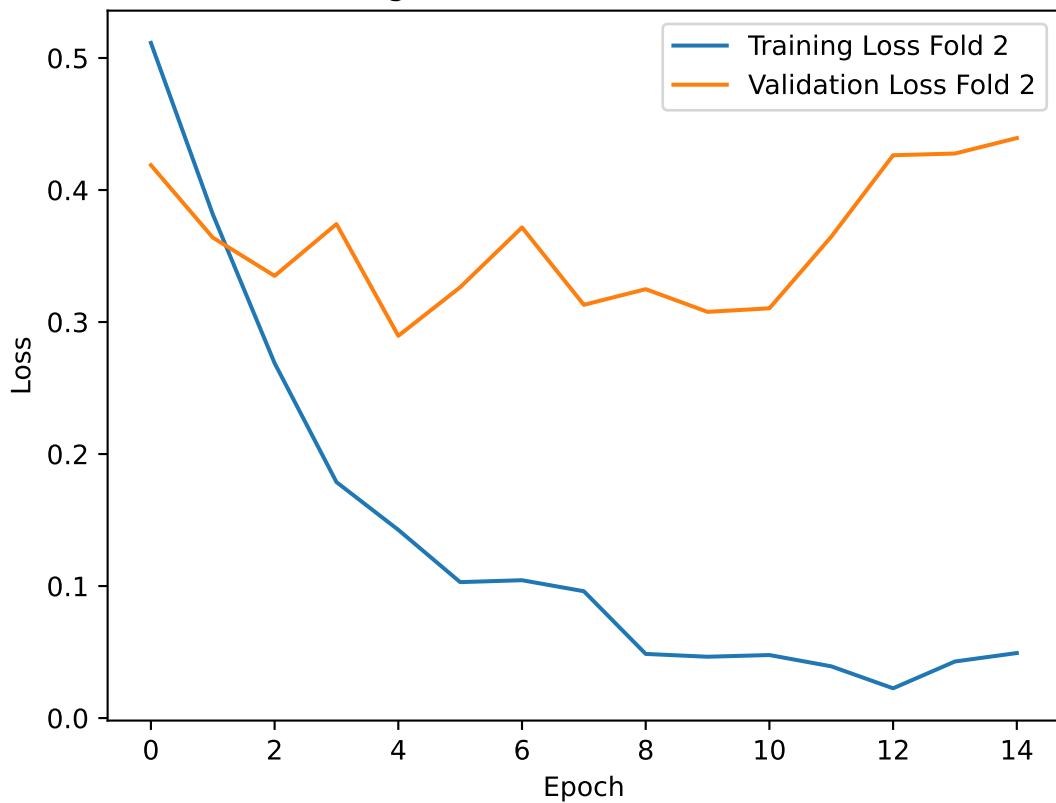
Downloading data from
https://storage.googleapis.com/tensorflow/keras-applications/resnet/resnet50_weights_tf_dim_ordering_tf_kernels_notop.h5



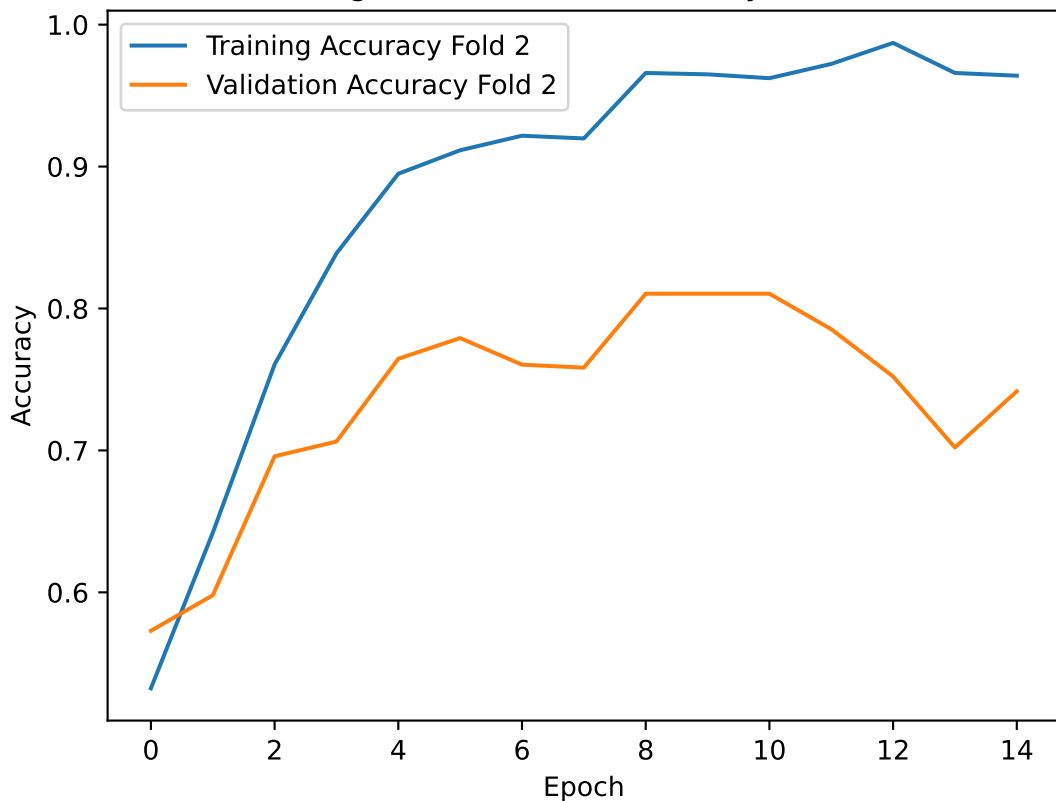
Validation F1-Score for fold 2: 0.058394160583941604

0.058394160583941604

Training and Validation Loss for Fold 2



Training and Validation Accuracy for Fold 2



Best Hyperparameters for Fold 2:

`{'conv_filters': 256, 'conv_activation': 'relu', 'unitsD1': 416, 'unitsD2': 160}`

Best Trial Score for Fold 2: 0.7895833253860474

Search: Running Trial #1

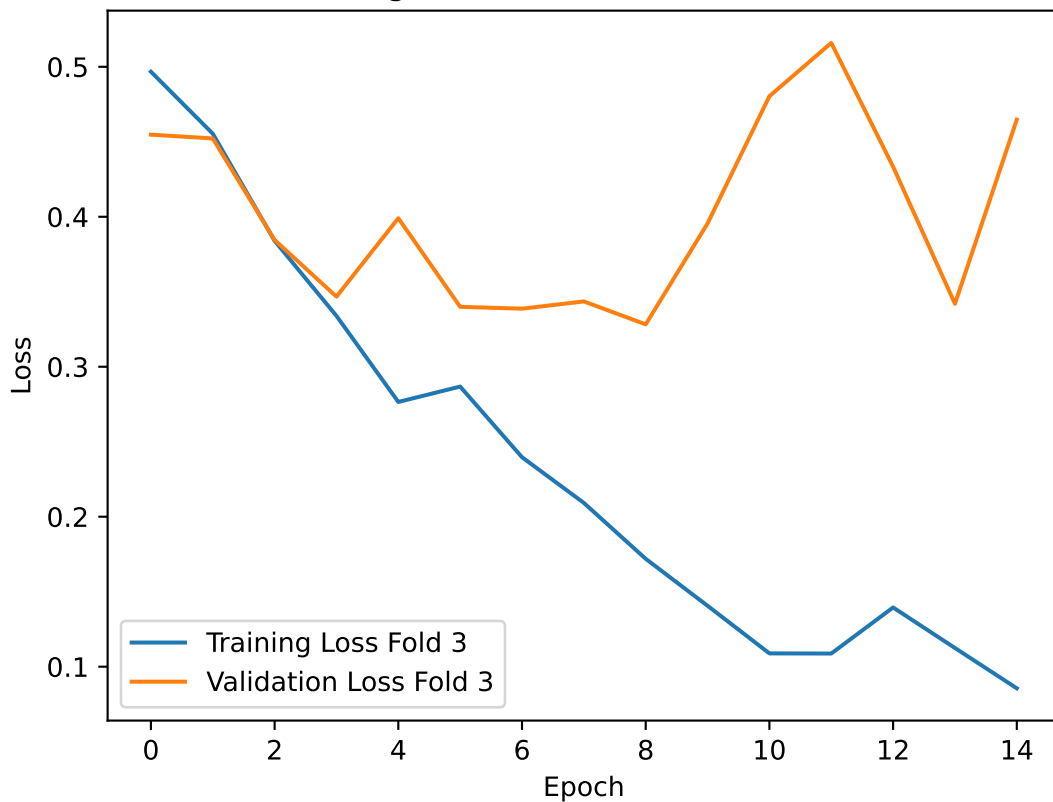
Value	Best Value So Far	Hyperparameter
224	224	conv_filters
relu	relu	conv_activation
288	288	unitsD1
32	32	unitsD2

Epoch 1/15
[0m 1/17][0m 32m[0m 37m[0m 1m9:53[0m 37s/step - accuracy: 0.5625 - loss: 0.4967
[0m 2/17][0m 32m[0m 37m[0m 1m3s[0m 261ms/step - accuracy: 0.5156 - loss: 0.5335
[0m 3/17][0m 32m[0m 37m[0m 1m3s[0m 260ms/step - accuracy: 0.4965 - loss: 0.5418
[0m 4/17][0m 32m[0m 37m[0m 1m3s[0m 261ms/step - accuracy: 0.4935 - loss: 0.5391
[0m 5/17][0m 32m[0m 37m[0m 1m3s[0m 263ms/step - accuracy: 0.4973 - loss: 0.5298
[0m 6/17][0m 32m[0m 37m[0m 1m2s[0m 264ms/step - accuracy: 0.4995 - loss: 0.5253
[0m 7/17][0m 32m[0m 37m[0m 1m2s[0m 263ms/step - accuracy: 0.5008 - loss: 0.5237
[0m 8/17][0m 32m[0m 37m[0m 1m2s[0m 263ms/step - accuracy: 0.5041 - loss: 0.5216
[0m 9/17][0m 32m[0m 37m[0m 1m2s[0m 263ms/step - accuracy: 0.5099 - loss: 0.5170
[0m 10/17][0m 32m[0m 37m[0m 1m1s[0m 264ms/step - accuracy: 0.5148 - loss: 0.5129
[0m 11/17][0m 32m[0m 37m[0m 1m1s[0m 264ms/step - accuracy: 0.5181 - loss: 0.5104
[0m 12/17][0m 32m[0m 37m[0m 1m1s[0m 264ms/step - accuracy: 0.5205 - loss: 0.5085
[0m 13/17][0m 32m[0m 37m[0m 1m1s[0m 264ms/step - accuracy: 0.5215 - loss: 0.5078
[0m 14/17][0m 32m[0m 37m[0m 1m0s[0m 264ms/step - accuracy: 0.5213 - loss: 0.5077
[0m 15/17][0m 32m[0m 37m[0m 1m0s[0m 264ms/step - accuracy: 0.5215 - loss: 0.5073
[0m 16/17][0m 32m[0m 37m[0m 1m0s[0m 265ms/step - accuracy: 0.5220 - loss: 0.5069
[0m 17/17][0m 32m[0m 37m[0m 1m0s[0m 842ms/step - accuracy: 0.5224 - loss: 0.5064
[0m 1m70s[0m 2s/step - accuracy: 0.5228 - loss: 0.5059 - val_accuracy: 0.5312 - val_loss: 0.4883
Epoch 2/15
[0m 1/17][0m 32m[0m 37m[0m 1m5s[0m 322ms/step - accuracy: 0.5625 - loss: 0.4567
[0m 2/17][0m 32m[0m 37m[0m 1m3s[0m 261ms/step - accuracy: 0.5000 - loss: 0.5457
[0m 3/17][0m 32m[0m 37m[0m 1m3s[0m 263ms/step - accuracy: 0.4618 - loss: 0.5672
[0m 4/17][0m 32m[0m 37m[0m 1m3s[0m 264ms/step - accuracy: 0.4479 - loss: 0.5702
[0m 5/17][0m 32m[0m 37m[0m 1m3s[0m 264ms/step - accuracy: 0.4521 - loss: 0.5610
[0m 6/17][0m 32m[0m 37m[0m 1m2s[0m 264ms/step - accuracy: 0.4557 - loss: 0.5563
[0m 7/17][0m 32m[0m 37m[0m 1m2s[0m 265ms/step - accuracy: 0.4589 - loss: 0.5519
[0m 8/17][0m 32m[0m 37m[0m 1m2s[0m 266ms/step - accuracy: 0.4669 - loss: 0.5464
[0m 9/17][0m 32m[0m 37m[0m 1m2s[0m 266ms/step - accuracy: 0.4764 - loss: 0.5397
[0m 10/17][0m 32m[0m 37m[0m 1m1s[0m 266ms/step - accuracy: 0.4856 - loss: 0.5323
[0m 11/17][0m 32m[0m 37m[0m 1m1s[0m 266ms/step - accuracy: 0.4942 - loss: 0.5258
[0m 12/17][0m 32m[0m 37m[0m 1m1s[0m 266ms/step - accuracy: 0.5027 - loss: 0.5199
[0m 13/17][0m 32m[0m 37m[0m 1m1s[0m 267ms/step - accuracy: 0.5099 - loss: 0.5150
[0m 14/17][0m 32m[0m 37m[0m 1m0s[0m 267ms/step - accuracy: 0.5154 - loss: 0.5113
[0m 15/17][0m 32m[0m 37m[0m 1m0s[0m 267ms/step - accuracy: 0.5209 - loss: 0.5072
[0m 16/17][0m 32m[0m 37m[0m 1m0s[0m 267ms/step - accuracy: 0.5262 - loss: 0.5031
[0m 17/17][0m 32m[0m 37m[0m 1m0s[0m 262ms/step - accuracy: 0.5304 - loss: 0.4999
[0m 1m7s[0m 408ms/step - accuracy: 0.5341 - loss: 0.4971 - val_accuracy: 0.6333 - val_loss: 0.4107
Epoch 3/15

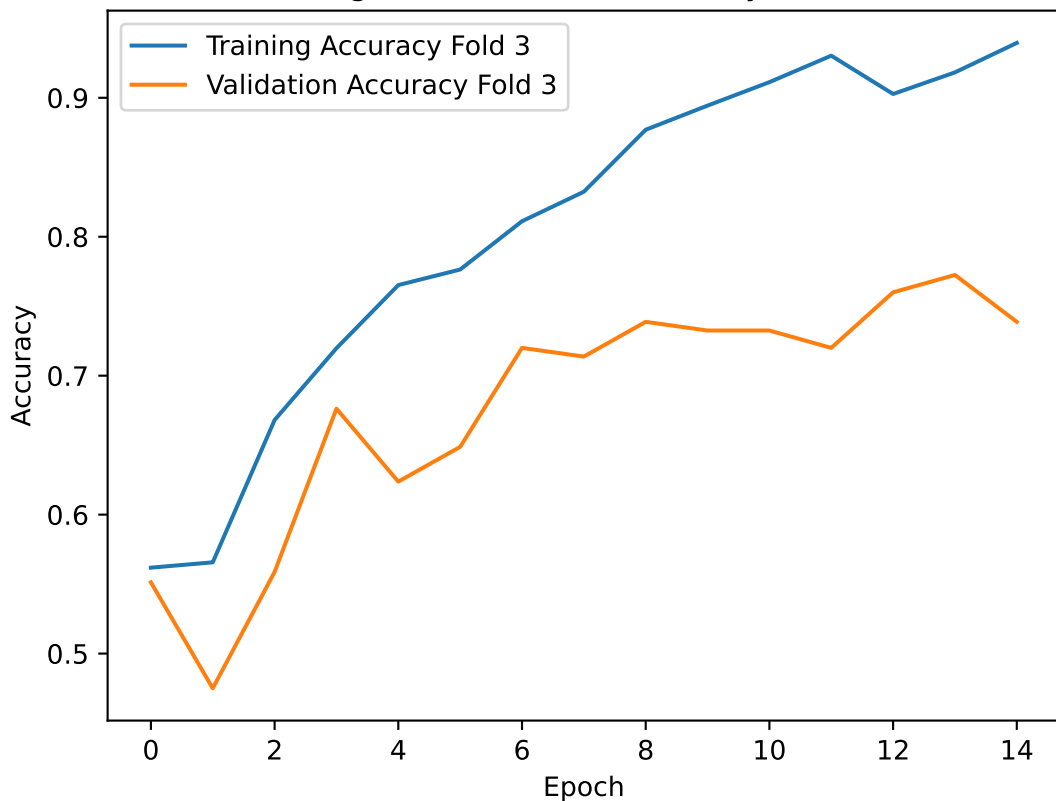
Validation F1-Score for fold 3: 0.1619718309859155

0.1619718309859155

Training and Validation Loss for Fold 3



Training and Validation Accuracy for Fold 3



Best Hyperparameters for Fold 3:

```
{'conv_filters': 192, 'conv_activation': 'relu', 'unitsD1': 32, 'unitsD2': 160}
```

Best Trial Score for Fold 3: 0.7787500023841858

Search: Running Trial #1

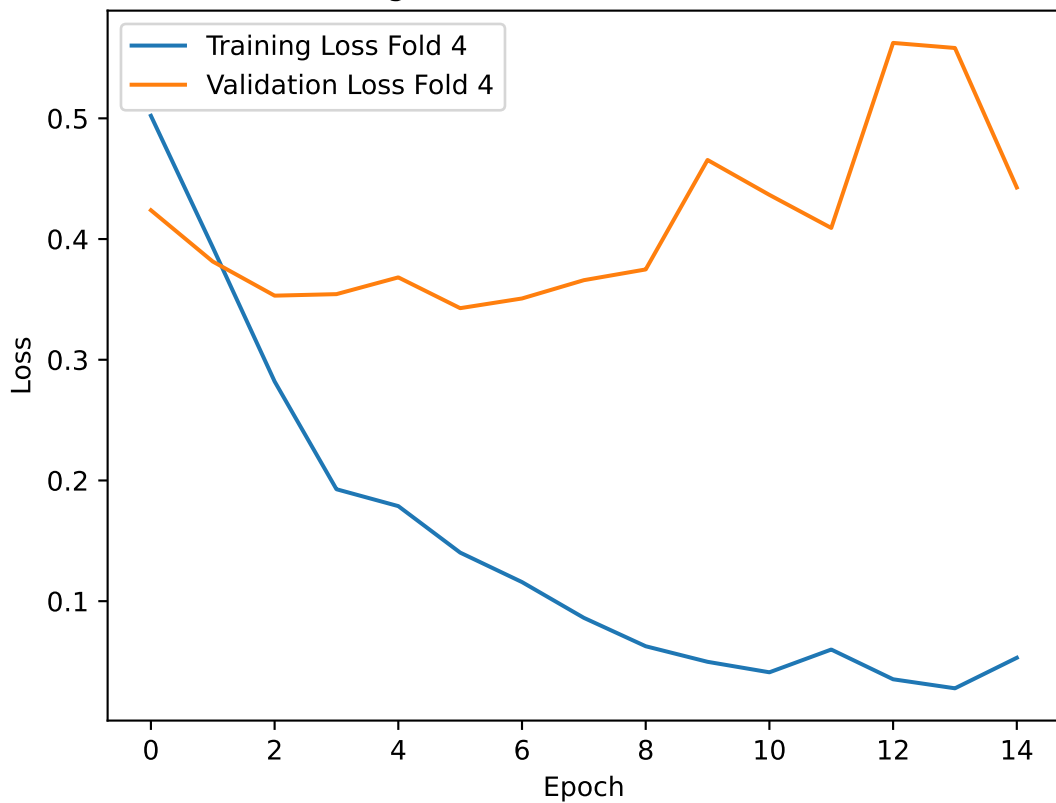
Value	Best Value So Far	Hyperparameter
192	192	conv_filters
relu	relu	conv_activation
32	32	unitsD1
160	160	unitsD2

Epoch 1/15
[0m 1/17][0m 32m-0m 37m-----0m 1m10:260m 39s/step - accuracy: 0.5312 - loss: 0.5212[0m 1m 2/17][0m 32m-0m 37m-----0m 1m3s0m 255ms/step - accuracy: 0.4766 - loss: 0.5205[0m 1m 3/17][0m 32m-0m 37m-----0m 1m3s0m 254ms/step - accuracy: 0.4705 - loss: 0.5183[0m 1m 4/17][0m 32m-0m 37m-----0m 1m3s0m 255ms/step - accuracy: 0.4681 - loss: 0.5194[0m 1m 5/17][0m 32m-0m 37m-----0m 1m3s0m 255ms/step - accuracy: 0.4745 - loss: 0.5152[0m 1m 6/17][0m 32m-0m 37m-----0m 1m2s0m 256ms/step - accuracy: 0.4796 - loss: 0.5126[0m 1m 7/17][0m 32m-0m 37m-----0m 1m2s0m 256ms/step - accuracy: 0.4812 - loss: 0.5144[0m 1m 8/17][0m 32m-0m 37m-----0m 1m2s0m 256ms/step - accuracy: 0.4846 - loss: 0.5148[0m 1m 9/17][0m 32m-0m 37m-----0m 1m2s0m 256ms/step - accuracy: 0.4917 - loss: 0.5134[0m 1m10/17][0m 32m-0m 37m-----0m 1m1s0m 256ms/step - accuracy: 0.4984 - loss: 0.5121[0m 1m11/17][0m 32m-0m 37m-----0m 1m1s0m 256ms/step - accuracy: 0.5045 - loss: 0.5111[0m 1m12/17][0m 32m-0m 37m-----0m 1m1s0m 256ms/step - accuracy: 0.5089 - loss: 0.5102[0m 1m13/17][0m 32m-0m 37m-----0m 1m1s0m 256ms/step - accuracy: 0.5112 - loss: 0.5097[0m 1m14/17][0m 32m-0m 37m-----0m 1m0s0m 256ms/step - accuracy: 0.5123 - loss: 0.5095[0m 1m15/17][0m 32m-0m 37m-----0m 1m0s0m 256ms/step - accuracy: 0.5133 - loss: 0.5093[0m 1m16/17][0m 32m-0m 37m-----0m 1m0s0m 256ms/step - accuracy: 0.5144 - loss: 0.5090[0m 1m17/17][0m 32m-0m 37m0m 1m0s0m 1s/step - accuracy: 0.5153 - loss: 0.5088[0m 1m17/17][0m 32m-0m 37m0m 1m76s0m 2s/step - accuracy: 0.5161 - loss: 0.5086 - val_accuracy: 0.6262 - val_loss: 0.4695
Epoch 2/15
[0m 1m 1/17][0m 32m-0m 37m-----0m 1m5s0m 324ms/step - accuracy: 0.6875 - loss: 0.4817[0m 1m 2/17][0m 32m-0m 37m-----0m 1m3s0m 258ms/step - accuracy: 0.6250 - loss: 0.4847[0m 1m 3/17][0m 32m-0m 37m-----0m 1m3s0m 258ms/step - accuracy: 0.6146 - loss: 0.4853[0m 1m 4/17][0m 32m-0m 37m-----0m 1m3s0m 259ms/step - accuracy: 0.5996 - loss: 0.4854[0m 1m 5/17][0m 32m-0m 37m-----0m 1m3s0m 261ms/step - accuracy: 0.5972 - loss: 0.4813[0m 1m 6/17][0m 32m-0m 37m-----0m 1m2s0m 261ms/step - accuracy: 0.5966 - loss: 0.4789[0m 1m 7/17][0m 32m-0m 37m-----0m 1m2s0m 261ms/step - accuracy: 0.5943 - loss: 0.4777[0m 1m 8/17][0m 32m-0m 37m-----0m 1m2s0m 260ms/step - accuracy: 0.5942 - loss: 0.4761[0m 1m 9/17][0m 32m-0m 37m-----0m 1m2s0m 260ms/step - accuracy: 0.5988 - loss: 0.4735[0m 1m10/17][0m 32m-0m 37m-----0m 1m1s0m 261ms/step - accuracy: 0.6036 - loss: 0.4711[0m 1m11/17][0m 32m-0m 37m-----0m 1m1s0m 261ms/step - accuracy: 0.6079 - loss: 0.4695[0m 1m12/17][0m 32m-0m 37m-----0m 1m1s0m 261ms/step - accuracy: 0.6108 - loss: 0.4680[0m 1m13/17][0m 32m-0m 37m-----0m 1m1s0m 261ms/step - accuracy: 0.6121 - loss: 0.4673[0m 1m14/17][0m 32m-0m 37m-----0m 1m0s0m 261ms/step - accuracy: 0.6125 - loss: 0.4672[0m 1m15/17][0m 32m-0m 37m-----0m 1m0s0m 261ms/step - accuracy: 0.6124 - loss: 0.4670[0m 1m16/17][0m 32m-0m 37m-----0m 1m0s0m 261ms/step - accuracy: 0.6131 - loss: 0.4667[0m 1m17/17][0m 32m-0m 37m0m 1m0s0m 257ms/step - accuracy: 0.6128 - loss: 0.4668[0m 1m17/17][0m 32m-0m 37m0m 1m5s0m 313ms/step - accuracy: 0.6125 - loss: 0.4669 - val_accuracy: 0.5800 - val_loss: 0.4243
Epoch 3/15

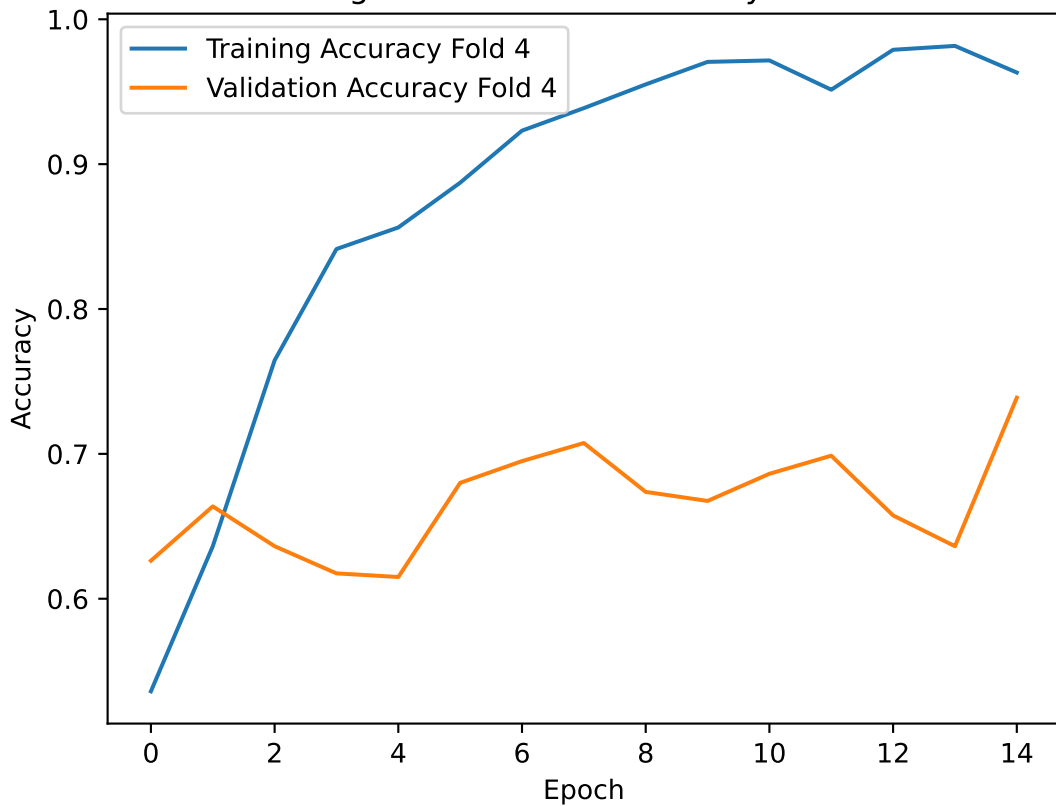
Validation F1-Score for fold 4: 0.036900369003690044

0.036900369003690044

Training and Validation Loss for Fold 4



Training and Validation Accuracy for Fold 4



Best Hyperparameters for Fold 4:

`{'conv_filters': 160, 'conv_activation': 'relu', 'unitsD1': 416, 'unitsD2': 160}`

Best Trial Score for Fold 4: 0.7849999666213989

Search: Running Trial #1

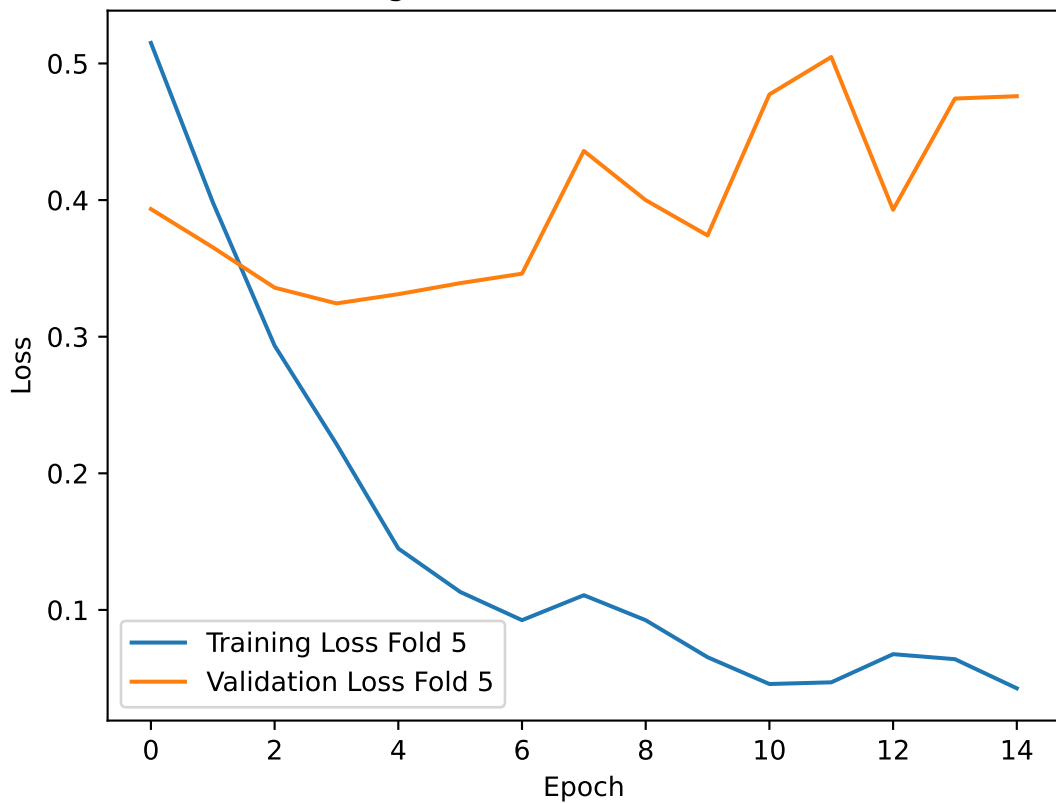
Value	Best Value So Far	Hyperparameter
192	192	conv_filters
relu	relu	conv_activation
416	416	unitsD1
160	160	unitsD2

Epoch 1/15
[1m 1/17][0m [32m][0m[37m[0m [1m9:46][0m 37s/step - accuracy: 0.6250 - loss: 0.5138
[32m][0m[37m[0m [1m4s][0m 272ms/step - accuracy: 0.5625 - loss: 0.5329
[32m][0m[37m[0m [1m3s][0m 268ms/step - accuracy: 0.5382 - loss: 0.5353
[32m][0m[37m[0m [1m3s][0m 267ms/step - accuracy: 0.5189 - loss: 0.5363
[32m][0m[37m[0m [1m3s][0m 269ms/step - accuracy: 0.5201 - loss: 0.5283
[32m][0m[37m[0m [1m2s][0m 269ms/step - accuracy: 0.5185 - loss: 0.5240
[32m][0m[37m[0m [1m2s][0m 269ms/step - accuracy: 0.5184 - loss: 0.5200
[32m][0m[37m[0m [1m2s][0m 269ms/step - accuracy: 0.5195 - loss: 0.5168
[32m][0m[37m[0m [1m2s][0m 269ms/step - accuracy: 0.5255 - loss: 0.5104
[32m][0m[37m[0m [1m1s][0m 270ms/step - accuracy: 0.5313 - loss: 0.5041
[32m][0m[37m[0m [1m1s][0m 270ms/step - accuracy: 0.5362 - loss: 0.4994
[32m][0m[37m[0m [1m1s][0m 270ms/step - accuracy: 0.5406 - loss: 0.4950
[32m][0m[37m[0m [1m1s][0m 270ms/step - accuracy: 0.5434 - loss: 0.4921
[32m][0m[37m[0m [1m0s][0m 270ms/step - accuracy: 0.5452 - loss: 0.4902
[32m][0m[37m[0m [1m0s][0m 270ms/step - accuracy: 0.5468 - loss: 0.4888
[32m][0m[37m[0m [1m0s][0m 270ms/step - accuracy: 0.5475 - loss: 0.4878
[32m][0m[37m[0m [1m0s][0m 273ms/step - accuracy: 0.5475 - loss: 0.4873
[32m][0m[37m[0m [1m69s][0m 2s/step - accuracy: 0.5475 - loss: 0.4868 - val_accuracy: 0.6325 - val_loss: 0.4077
Epoch 2/15
[1m 1/17][0m [32m][0m[37m[0m [1m5s][0m 358ms/step - accuracy: 0.7188 - loss: 0.3957
[32m][0m[37m[0m [1m4s][0m 268ms/step - accuracy: 0.6562 - loss: 0.4345
[32m][0m[37m[0m [1m3s][0m 269ms/step - accuracy: 0.6319 - loss: 0.4429
[32m][0m[37m[0m [1m3s][0m 270ms/step - accuracy: 0.6126 - loss: 0.4503
[32m][0m[37m[0m [1m3s][0m 271ms/step - accuracy: 0.6114 - loss: 0.4440
[32m][0m[37m[0m [1m2s][0m 272ms/step - accuracy: 0.6102 - loss: 0.4426
[32m][0m[37m[0m [1m2s][0m 271ms/step - accuracy: 0.6110 - loss: 0.4405
[32m][0m[37m[0m [1m2s][0m 271ms/step - accuracy: 0.6152 - loss: 0.4369
[32m][0m[37m[0m [1m2s][0m 272ms/step - accuracy: 0.6228 - loss: 0.4295
[32m][0m[37m[0m [1m1s][0m 273ms/step - accuracy: 0.6302 - loss: 0.4213
[32m][0m[37m[0m [1m1s][0m 273ms/step - accuracy: 0.6360 - loss: 0.4150
[32m][0m[37m[0m [1m1s][0m 273ms/step - accuracy: 0.6409 - loss: 0.4096
[32m][0m[37m[0m [1m1s][0m 273ms/step - accuracy: 0.6454 - loss: 0.4055
[32m][0m[37m[0m [1m0s][0m 273ms/step - accuracy: 0.6486 - loss: 0.4027
[32m][0m[37m[0m [1m0s][0m 273ms/step - accuracy: 0.6517 - loss: 0.3999
[32m][0m[37m[0m [1m0s][0m 273ms/step - accuracy: 0.6543 - loss: 0.3975
[32m][0m[37m[0m [1m0s][0m 269ms/step - accuracy: 0.6556 - loss: 0.3958
[32m][0m[37m[0m [1m7s][0m 438ms/step - accuracy: 0.6568 - loss: 0.3944 - val_accuracy: 0.6825 - val_loss: 0.3780
Epoch 3/15

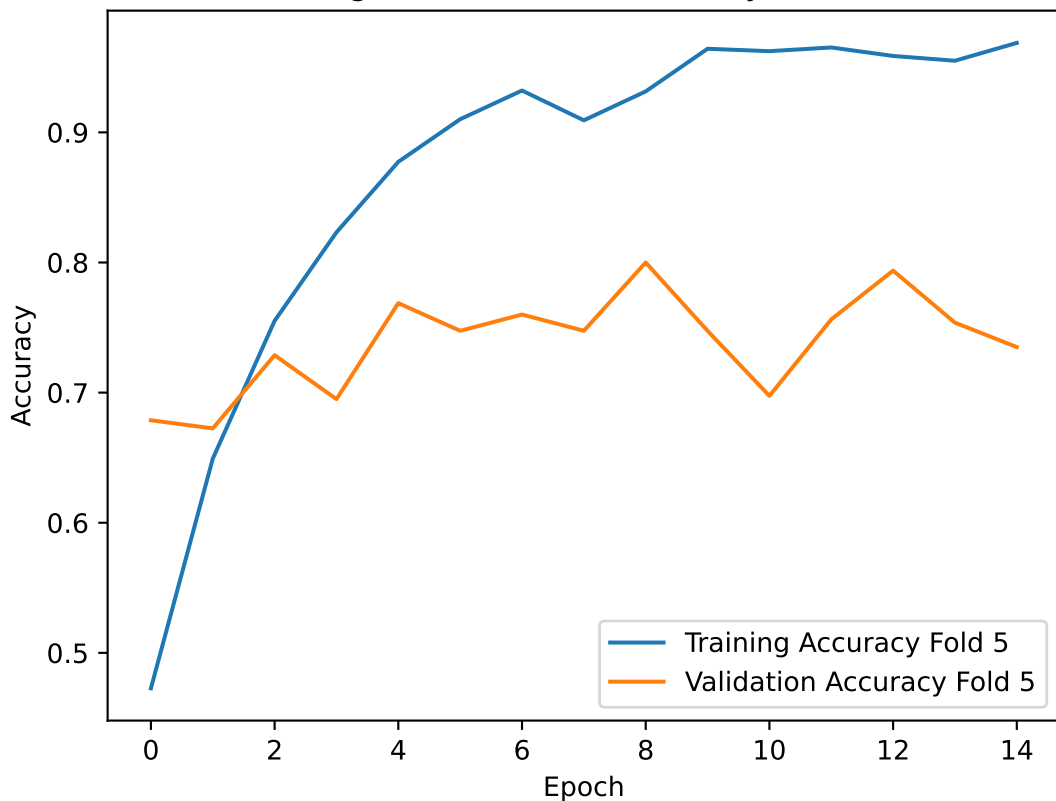
Validation F1-Score for fold 5: 0.1423487544483986

0.1423487544483986

Training and Validation Loss for Fold 5



Training and Validation Accuracy for Fold 5



Best Hyperparameters for Fold 5:

`{'conv_filters': 64, 'conv_activation': 'relu', 'unitsD1': 288, 'unitsD2': 32}`

Best Trial Score for Fold 5: 0.7787500023841858

Search: Running Trial #1

Value	Best Value So Far	Hyperparameter
256	256	conv_filters
relu	relu	conv_activation
32	32	unitsD1
32	32	unitsD2

Epoch 1/15
[1m 1/17][0m [32m][0m][37m][0m [1m10:13][0m 38s/step - accuracy: 0.5938 - loss: 0.4905][1m 2/17][0m [32m][0m][37m][0m [1m3s][0m 255ms/step - accuracy: 0.5078 - loss: 0.5064][1m 3/17][0m [32m][0m][37m][0m [1m3s][0m 256ms/step - accuracy: 0.4635 - loss: 0.5108][1m 4/17][0m [32m][0m][37m][0m [1m3s][0m 255ms/step - accuracy: 0.4473 - loss: 0.5095][1m 5/17][0m [32m][0m][37m][0m [1m3s][0m 257ms/step - accuracy: 0.4453 - loss: 0.5050][1m 6/17][0m [32m][0m][37m][0m [1m2s][0m 257ms/step - accuracy: 0.4466 - loss: 0.5036][1m 7/17][0m [32m][0m][37m][0m [1m2s][0m 257ms/step - accuracy: 0.4491 - loss: 0.5022][1m 8/17][0m [32m][0m][37m][0m [1m2s][0m 257ms/step - accuracy: 0.4535 - loss: 0.5009][1m 9/17][0m [32m][0m][37m][0m [1m2s][0m 257ms/step - accuracy: 0.4610 - loss: 0.4979][1m10/17][0m [32m][0m][37m][0m [1m1s][0m 257ms/step - accuracy: 0.4668 - loss: 0.4956][1m11/17][0m [32m][0m][37m][0m [1m1s][0m 257ms/step - accuracy: 0.4719 - loss: 0.4947][1m12/17][0m [32m][0m][37m][0m [1m1s][0m 257ms/step - accuracy: 0.4760 - loss: 0.4939][1m13/17][0m [32m][0m][37m][0m [1m1s][0m 257ms/step - accuracy: 0.4784 - loss: 0.4947][1m14/17][0m [32m][0m][37m][0m [1m0s][0m 257ms/step - accuracy: 0.4799 - loss: 0.4960][1m15/17][0m [32m][0m][37m][0m [1m0s][0m 257ms/step - accuracy: 0.4822 - loss: 0.4963][1m16/17][0m [32m][0m][37m][0m [1m0s][0m 257ms/step - accuracy: 0.4846 - loss: 0.4964][1m17/17][0m [32m][0m][37m][0m [1m0s][0m 853ms/step - accuracy: 0.4865 - loss: 0.4966][1m17/17][0m [32m][0m][37m][0m [1m71s][0m 2s/step - accuracy: 0.4883 - loss: 0.4968 - val_accuracy: 0.6350 - val_loss: 0.5207
Epoch 2/15
[1m 1/17][0m [32m][0m][37m][0m [1m5s][0m 326ms/step - accuracy: 0.5312 - loss: 0.6312][1m 2/17][0m [32m][0m][37m][0m [1m3s][0m 254ms/step - accuracy: 0.5234 - loss: 0.6755][1m 3/17][0m [32m][0m][37m][0m [1m3s][0m 255ms/step - accuracy: 0.5260 - loss: 0.6650][1m 4/17][0m [32m][0m][37m][0m [1m3s][0m 255ms/step - accuracy: 0.5332 - loss: 0.6435][1m 5/17][0m [32m][0m][37m][0m [1m3s][0m 257ms/step - accuracy: 0.5403 - loss: 0.6242][1m 6/17][0m [32m][0m][37m][0m [1m2s][0m 257ms/step - accuracy: 0.5414 - loss: 0.6107][1m 7/17][0m [32m][0m][37m][0m [1m2s][0m 257ms/step - accuracy: 0.5457 - loss: 0.5979][1m 8/17][0m [32m][0m][37m][0m [1m2s][0m 257ms/step - accuracy: 0.5532 - loss: 0.5835][1m 9/17][0m [32m][0m][37m][0m [1m2s][0m 257ms/step - accuracy: 0.5631 - loss: 0.5678][1m10/17][0m [32m][0m][37m][0m [1m1s][0m 257ms/step - accuracy: 0.5715 - loss: 0.5535][1m11/17][0m [32m][0m][37m][0m [1m1s][0m 257ms/step - accuracy: 0.5774 - loss: 0.5429][1m12/17][0m [32m][0m][37m][0m [1m1s][0m 257ms/step - accuracy: 0.5831 - loss: 0.5334][1m13/17][0m [32m][0m][37m][0m [1m1s][0m 257ms/step - accuracy: 0.5867 - loss: 0.5260][1m14/17][0m [32m][0m][37m][0m [1m0s][0m 257ms/step - accuracy: 0.5889 - loss: 0.5200][1m15/17][0m [32m][0m][37m][0m [1m0s][0m 257ms/step - accuracy: 0.5913 - loss: 0.5142][1m16/17][0m [32m][0m][37m][0m [1m0s][0m 257ms/step - accuracy: 0.5936 - loss: 0.5090][1m17/17][0m [32m][0m][37m][0m [1m0s][0m 252ms/step - accuracy: 0.5952 - loss: 0.5045][1m17/17][0m [32m][0m][37m][0m [1m6s][0m 350ms/step - accuracy: 0.5967 - loss: 0.5006 - val_accuracy: 0.6662 - val_loss: 0.4369
Epoch 3/15

Average Validation F1-Score across all folds:

0.11254713647956645

Average Validation MSE across all folds:

0.38412981986999506

Average Validation Accuracy across all folds:

0.7073222251733144

Best Hyperparameters Found:

Best Fold: 1

Best Trial Score: 0.8166667222976685

Best Hyperparameters: {'conv_filters': 64, 'conv_activation': 'relu', 'unitsD